



Image watermarking using polar harmonic transform with parameters in $SL(2, \mathbb{R})$



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ABSTRACT

In this paper, a novel polar harmonic transform with parameters in $SL(2, \mathbb{R})$ is proposed in order to increase the security of the watermark information. It is named as the Polar Linear Canonical Transform (PLCT) based on the linear canonical transform. Moreover, a watermarking scheme associated with the PLCT is presented, and the simulations are performed to verify the importance of parameters of the PLCT and improve the capabilities of the existing results on the watermarking information's security. The experimental results show that due to the varieties of parameters, if parameters chosen in the extraction process are inconsistent with those used in the embedding process, then the bit error rate is bigger which means it is worse than that of consistent parameters in the two processes. This confirms the fact that parameters of the PLCT have superior security for watermark information.

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1. Introduction

Nowadays digital images have become a significant part in our life with the development of science and technology; to protect people's information, the security and privacy of digital images become a cutting issue which has been drawing many researchers' attention in recent years. Thus, the information security has turned to be an important research field. As a main branch of the information security, the watermark is recognized for its remarkable advancement, because it guarantees the security and privacy of digital images.

The watermark techniques play an essential role in the watermark. Among them, the invariant moments and transforms are hot topics, which have been successfully applied in watermark, as well as pattern recognition, character recognition, face recognition [2]. Alghoniem and Tewfik [1] first

proposed the geometric moments of images for the watermarking and proved that the techniques had high robustness to malicious attacks. Subsequently, several moments and transforms, such as the Zernike Moments (ZMs), the Pseudo-Zernike Moments (PZMs), the Tchebichef Moments (TMs), the Complex Moments (CMs), and the Polar Harmonic Transforms (PHTs), have been used in image watermarking [4–7,9–12]. Thereinto, the PHTs, including the Polar Complex Exponential Transform (PCET), the Polar Cosine Transform (PCT) and the Polar Sine Transform (PST), were introduced by Yap et al. [3] and accepted for their less computation and less numerical stability issue compared with the ZMs, the PZMs and the CMs. Since then, the PHTs were successfully applied in image watermarking, such as [10–12].

The LCT introduced during 1970s [13] is a linear integral transform with parameters and has been proved to be a powerful tool in several areas, such as optics, optimal filtering, signal processing, and time–frequency analysis [14,16–22]. By choosing suitable parameters, the LCT can be reduced to particular cases, such as the classical Fourier transform (FT), the fractional Fourier transform (FrFT), and the Fresnel transform (FRT). At present, the well known

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concepts, such as the uncertainty principles, the convolution and product theorem, and the Hilbert-transform have been well investigated and extended in the LCT domain. To enhance the security of watermark information, we combine the moments and transforms with the Linear Canonical Transform (LCT).

In this paper, a novel polar harmonic transform, named as the Polar Linear Canonical Transform (PLCT), with parameters in the real special linear group $SL(2, \mathbb{R})$ is proposed, which is applied in the image watermarking. The PLCT is the generalization of the traditional PCET. Subsequently, under the scheme of the watermark [9], the simulations confirm stronger capabilities of the PLCT for many varieties of their parameters compared with the PCET. In other words, parameters of the novel transform can influence the bit error rate of the extracted watermark bits when parameters are inconsistent in the watermark embedding and in the watermark extraction. Therefore, this consistency of parameters can improve the safety of the watermark information.

The remainder of this paper is organized as follows. In Section 2, we present some basic concepts, and propose the novel PLCT. In Section 3, we investigate the accurate transform coefficients of the novel transform. Combining with the novel transforms, the watermark embedding and the watermark extraction are introduced in Sections 4 and 5 respectively. In Section 6, the simulation results illustrate performances of the PLCT on numerical stability, watermark invisibility, watermark capacity, watermark robustness, and the influence of parameter sensitivity on the watermark extraction process. Discussions and conclusions are given in the last.

2. Novel polar harmonic transform based on the LCT

2.1. Moments and transforms

In this subsection, we briefly present some basic concepts about moments, transforms and the theory of the LCT before introducing the newly transform associated with the LCT.

Let f be an image function. Then its moments or transform coefficients M_{nm} are defined as

$$M_{nm} = \iint_{x^2+y^2 \leq 1} K_{nm}^*(x, y) f(x, y) dx dy \\ = \int_0^{2\pi} \int_0^1 K_{nm}^*(r, \theta) f(r, \theta) r dr d\theta, \quad (1)$$

with

$$K_{nm}(r, \theta) = R_n(r) A_m(\theta),$$

where $*$ is denoted as the complex conjugation operation, $R_n(r)$ and $A_m(\theta)$ are respectively the radial component and the angle component, and n and m are both integers. If the kernel K_{nm} satisfies the orthogonality condition

$$\int_0^{2\pi} \int_0^1 K_{nm}(r, \theta) K_{nm'}^*(r, \theta) r dr d\theta \\ = \int_0^1 R_n(r) R_{n'}^*(r) r dr \int_0^{2\pi} A_m(\theta) A_{m'}^*(\theta) d\theta = \delta_{nn'} \delta_{mm'},$$

where c_1 and c_2 are constants,

$$\delta_{nn'} = \begin{cases} 1 & \text{if } n = n' \\ 0 & \text{if } n \neq n' \end{cases}$$

and

$$\delta_{mm'} = \begin{cases} 1 & \text{if } m = m' \\ 0 & \text{if } m \neq m', \end{cases}$$

then the method with K_{nm} is orthogonal. Thus, the image function $f(x, y)$ is represented as

$$f(x, y) = \sum_n \sum_m M_{nm} K_{nm}(x, y). \quad (2)$$

More important, we have obtained different moments or transforms by defining different functions $R_n(r)$. For example, the kernel of the PCET [3] is

$$K_{nm}^e(r, \theta) = R_n(r) A_m(\theta) \\ = \exp(i2\pi nr^2) \exp(im\theta), \quad (3)$$

the kernel of the ZMs and the kernel of the PZMs can be respectively defined as

$$K_{nm}^Z = \sum_{s=0}^{(n-|m|)/2} \frac{(-1)^s (n-s)! r^{n-2s}}{s! ((n-|m|)/2 - s)! (n-|m|-s)!} e^{-im\theta} \quad (n=0, 1, \dots)$$

and

$$K_{nm}^{PZ} = \sum_{s=0}^{n-|m|} \frac{(-1)^s (2n+1-s)! r^{n-s}}{s! (n-|m|-s)! (n-|m|-s)!} e^{-im\theta} \quad (n=0, 1, \dots)$$

where $i^2 = -1$.

Next, some important knowledge of the LCT needs to be presented as follows. Refer to [15,16] for more details about the LCT theory and its properties. For the sake of convenience, it is necessary to define some notations. A real special linear group is denoted as

$$SL(2, \mathbb{R}) = \{A \in M(2, \mathbb{R}) | \det(A) = 1\},$$

where $M(2, \mathbb{R})$ is a set of 2×2 real matrices, and $\mathbb{R}$ is real. In $SL(2, \mathbb{R})$, the distance of two matrices is measured by the Frobenius inner product defined by

$$\mathbf{d}(A, B) = \sqrt{\text{tr}((A-B)^T (A-B))}, \quad (4)$$

where $\text{tr}(\cdot)$ and the superscript T stand for the trace and the transpose, respectively.

The one-dimensional LCT (1D LCT) with parameter $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SL(2, \mathbb{R})$ of a function $f(x)$ is given by

$$L_f^A(u) = \mathcal{L}_A[f](u) \\ = \begin{cases} \int_{\mathbb{R}} f(x) K(u, x) dx, & b \neq 0, \\ \sqrt{d} \exp\left(-\frac{i}{2} c d u^2\right) f(du), & b = 0, \end{cases}$$

where

$$K(u, x) = \sqrt{\frac{1}{i2\pi b}} \exp\left\{\frac{i}{2b} [du^2 - 2ux + ax^2]\right\},$$

$\mathcal{L}_A$ is the unitary LCT operator, and L_f^A is the LCT of f .

In the following, we always write the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ as $A = [ab; cd]$ for convenience. Inspired by the methods of the moments and transforms, we propose a novel polar harmonic transforms associated with the LCT in the next subsection.

2.2. The definition of the PLCT

We define the Polar Linear Canonical Transform (PLCT) with $A = [ab; cd] \in SL(2, \mathbb{R})$ as

$$M_{nm}^{\text{PLCT}} = \frac{1}{\pi} \int_0^{2\pi} \int_0^1 K_{nm}^{A*}(r, \theta) f(r, \theta) r \, dr \, d\theta, \quad (5)$$

with

$$\begin{aligned} K_{nm}^A(r, \theta) &= R_n^A(r) \exp(im\theta) \\ &= \exp\left\{\frac{-i}{2b} [d(2\pi nb)^2 - 4r^2 \pi nb + ar^4]\right\} \exp(im\theta). \end{aligned} \quad (6)$$

It is worthy noting that the element $(1, 2)$ -th of A is not equal to zero, otherwise the transform is no sense. Since

$$K_{nm}^{A*}(r, \theta) = \exp\left\{\frac{i}{2b} [d(2\pi nb)^2 - 4r^2 \pi nb + ar^4]\right\} \exp(-im\theta),$$

thus, $|K_{nm}^A| \neq |K_{-n, -m}^A|$ and $|M_{nm}^A| \neq |M_{-n, -m}^A|$ if $A \neq [0b; -1/b0]$. It implies that the transform coefficients based on the PLCT with $A = [ab; cd] \in SL(2, \mathbb{R})$ have independent magnitudes, except for $A \neq [0b; -1/b0]$; otherwise the transform coefficients based on the PLCT have dependent magnitudes. Moreover, it has

$$\int_0^1 R_n^A(r) R_{n'}^{A*}(r) r \, dr = \frac{1}{2} \delta_{nn'},$$

and

$$\int_0^{2\pi} \int_0^1 K_{nm}^A(r, \theta) K_{n'm'}^{A*}(r, \theta) r \, dr \, d\theta = \pi \delta_{nn'} \delta_{mm'}.$$

Thus the kernel K_{nm}^A of the PLCT satisfies the orthogonality condition, and then the PLCT is an orthogonal transform. Furthermore, we know that the series $\{\exp(ik\pi t/T), |k| = 0, 1, \dots, \infty\}$ is a completely orthogonal basis on $t \in [0, 2T]$. Hence, the kernel K_{nm}^A of the PLCT is complete at $[0, 1] \times [0, 2\pi]$. Therefore, the kernel K_{nm}^A of the PLCT is orthogonal and complete at $[0, 1] \times [0, 2\pi]$. Meanwhile, since the kernel of the PLCT is a basic wave, we name it harmonic transform. It is worth to point out that the traditional PCET can be seen as a special case of the PLCT when choosing the parameter $A = [0b; -1/b0]$. In addition, the kernel of the PLCT has the form $\exp\{(i/2b)[d(2\pi nb)^2 + ar^4]\}$ which can be viewed as the offset form and may affect the radial component and the angle component, especially the factor a/b . In the simulation section, we need to examine the capabilities of the PLCT on image reconstruction, image watermarking, and analyze the effect of the parameter of the PLCT via simulations in this paper.

2.3. Discrete implementation of the PLCT

In this section, we need to consider the discrete implementation of the PLCT from the experimental view.

For any finite image function $f(x, y)$, its discrete form is written as $g[k, l]$ with the size $N \times M$ ($k = 0, 1, \dots, N-1, l = 0, 1, \dots, M-1$), and then $f(x, y)$ can be projected into a

domain $(x_k, y_l) \in [-1, 1] \times [-1, 1]$ with

$$x_k = \frac{k-N/2}{N/2}, \quad y_l = \frac{l-M/2}{M/2}.$$

In Cartesian coordinates, the discrete form of the PLCT is written as

$$M_{nm}^{\text{PLCT}} = \frac{1}{\pi} \sum_{l=0}^{M-1} \sum_{k=0}^{N-1} \tilde{K}_{nm}^{A*}(x_k, y_l) f(x_k, y_l) \Delta x \Delta y, \quad (7)$$

subject to $x_k^2 + y_l^2 \leq 1$, where $\tilde{K}_{nm}^A$ is determined by Eq. (6), $f(x_k, y_l) = g[k, l]$, $\Delta x = 2/N$, and $\Delta y = 2/M$.

From Eq. (2), the approximation of f based on the PLCT is given as follows:

$$\hat{f}(x, y) = \sum_S M_{nm}^{\text{PLCT}} \tilde{K}_{nm}^A(x, y), \quad (8)$$

where S is a subset of $l = \{(n, m) | n, m = 0, \pm 1, \pm 2, \dots, \pm \infty\}$.

3. Selection of accurate transform coefficients

We have obtained the transform coefficients from Eq. (7), however, the integral approximation error is inevitable because of the computation from the continuous domain to the discrete domain. For instance, let $f \equiv C$ (C is constant), then the transform coefficients of the PLCT are rewritten as

$$M_{nm}^{\text{PLCT}} = \begin{cases} 2C \int_0^1 \exp\left\{\frac{-i}{2b} [d(2\pi nb)^2 - 4r^2 \pi nb + ar^4]\right\} r \, dr, & m = 0, \\ 0, & m \neq 0, \end{cases}$$

where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SL(2, \mathbb{R})$.

However, in fact, we have the following theorem.

Theorem 3.1. Let $f(x, y)$ be a constant image function C , then the transform coefficients of f based on the PLCT are

$$M_{nm}^{\text{PLCT}} = \begin{cases} \text{nonzero}, & n = m = 0 \text{ or } n \neq 0, m = 0, \\ \text{nonzero}, & (n, m) \neq (0, 0), m = 4l, \\ 0 & \text{others}, \end{cases}$$

where l is an integer.

Proof. Under the first condition, the conclusion is obvious. From Eqs. (5) and (7), let $r_{kl} = \sqrt{x_k^2 + y_l^2}$ and $A = [ab; cd] \in SL(2, \mathbb{R})$, then $\theta_{kl} = \arccos(x_k/r_{kl})$, if $r_{kl} \neq 0$; otherwise, $\theta_{kl} = 0$.

On the one hand, taking $T_1 = \{(r_{kl}, \theta_{kl}) | (r_{kl}, \theta_{kl}) \in [0, 1] \times [0, \pi]\}$, we have

$$\begin{aligned} M_{nm}^{\text{PLCT}} &= \frac{4}{\pi NM} \sum_k \sum_l C K_{nm}^{A*}(r_{kl}, \theta_{kl}) \\ &= \frac{4C}{\pi NM} \sum_{k,l,T_1} \left[\exp\left\{\frac{-i}{2b} [d(2\pi nb)^2 - 4r_{kl}^2 \pi nb + ar_{kl}^4]\right\} e^{im\theta_{kl}} (1 + e^{im\pi}) \right]^*, \end{aligned}$$

and if $m = 2l + 1$ (l is integer), then $M_{nm}^{\text{PLCT}} = 0$.

By taking $T_2 = \{(r_{kl}, \theta_{kl}) | (r_{kl}, \theta_{kl}) \in [0, 1] \times ([0, \pi/2] \cup [\pi, 3\pi/2])\}$, we have

$$M_{nm}^{\text{PLCT}} = \frac{4}{\pi NM} \sum_k \sum_l C K_{nm}^{A*}(r_{kl}, \theta_{kl})$$

$$= \frac{4C}{\pi NM} \sum_{k,l,T_2} \left[\exp \left\{ \frac{-i}{2b} \left[d(2\pi nb)^2 - 4r_{kl}^2 \pi nb + ar_{kl}^4 \right] \right\} e^{im\theta_{kl}} (1 + e^{im\pi/2}) \right]^*$$

and if $m = 4l$ (l is integer), then $M_{nm}^{\text{PLCT}} \neq 0$; otherwise, $M_{nm}^{\text{PLCT}} = 0$. $\square$

According to the above analysis, we should use the accurate transform coefficients in image watermarking to avoid the computational error caused by the inaccurate coefficients. Hence, we need to fix the set of accurate transform coefficients and its total number which can be used in watermarking. From Eq. (8), the index (n, m) of the transform coefficients are restricted to the condition [3] $S = \{(n, m) | |n| + |m| \leq K\}$, the order K is an integer for a fair evaluation. It implies that the order K determines the total number of the transform coefficients used in watermarking. Therefore, for a given order K , the number of the transforms coefficients of the PLCT is $(K+1)^2 + K^2$. Moreover, the set of the final transform coefficients of the PLCT used in image watermarking is that

$$S_{\text{PLCT}}^1 = \{M_{nm}^{\text{PLCT}} | |n| + |m| \leq K, m \neq 4l\}, \quad (9)$$

having

$$\|S_{\text{PLCT}}^1\| = \begin{cases} \frac{3K^2 + 3K}{2}, & K = 4l \text{ or } K = 4l + 3, \\ \frac{3K^2 + 3K - 2}{2}, & K = 4l + 1 \text{ or } K = 4l + 2, \end{cases}$$

where $A \neq [0b; -1/b0]$, and l is an integer; and

$$S_{\text{PLCT}}^2 = \{M_{nm}^{\text{PLCT}} | |n| + |m| \leq K, n \geq 0, m \neq 4l\}, \quad (10)$$

with

$$\|S_{\text{PLCT}}^2\| = \begin{cases} \frac{3K^2 + 3K}{4}, & K = 4l \text{ or } K = 4l + 3, \\ \frac{3K^2 + 3K - 2}{4}, & K = 4l + 1 \text{ or } K = 4l + 2, \end{cases}$$

where $A = [0b; -1/b0]$ and l is an integer.

4. Watermark embedding

In this section, we introduce the watermark embedding process associating with the PLCT. Let $W = \{w_1, w_2, \dots, w_{N_{\max}}\} | w_p \in \{0, 1\}$ be a watermark sequence with the finite length $N_{\max}$. Then we can embed the watermark bit in W on an accurate transform coefficient based on the PLCT via the watermarking scheme [9]. More details of this watermarking scheme can be referred to [9].

Choosing the selected accurate transform coefficients $\{M_{n_p, m_p}^{\text{PLCT}} | p = 1, 2, \dots, N_{\max}\}$, we embed the watermark bit w_p of W on the $M_{n_p, m_p}^{\text{PLCT}}$ to obtain the dither quantization $\hat{M}_{n_p, m_p}^{\text{PLCT}}$ defined as

$$\hat{M}_{n_p, m_p}^{\text{PLCT}} = \left\lfloor \frac{|M_{n_p, m_p}^{\text{PLCT}}| - d_p(w_p)}{\mu} \right\rfloor \mu + d_p(w_p), \quad (11)$$

and

$$\hat{M}_{n_p, m_p}^{\text{PLCT}} = \frac{|\hat{M}_{n_p, m_p}^{\text{PLCT}}|}{|M_{n_p, m_p}^{\text{PLCT}}|} M_{n_p, m_p}^{\text{PLCT}}, \quad (12)$$

where $\lfloor \cdot \rfloor$ and $d(\cdot)$ denote respectively the round operation and a dither function, μ is a quantization step, $d_p(0) \in [0, \mu/2]$, and $d_p(1) = d_p(0) + \mu/2$.

Subsequently, to reconstruct a real image, we define the compensation as

$$\varepsilon(x, y) = \sum_{p=1}^{N_{\max}} 2 \operatorname{Re}((\hat{M}_{n_p, m_p}^{\text{PLCT}} - M_{n_p, m_p}^{\text{PLCT}}) K_{n_p, m_p}^A(x, y)), \quad (13)$$

where $\operatorname{Re}(\cdot)$ is denoted as the real part of a complex.

As a consequence, the watermarking image can be obtained by embedding the compensation image on the original image, i.e.,

$$\hat{f}_w(x, y) = f(x, y) + \varepsilon(x, y).$$

5. Watermark extraction

Next, we present the watermark extraction scheme [9] to extract watermark bits from the specially selected transform coefficients of $\hat{f}_w$ based on the PLCT. The selection of transform coefficients in extraction process is the same as the selection in watermarking embedding. Thus, the correct transform coefficients $\{\bar{M}_{n_p, m_p}^{\text{PLCT}} | p = 1, 2, \dots, N_{\max}\}$ of a watermarking image $\hat{f}_w$ can be selected. Then the quantized value of $\bar{M}_{n_p, m_p}^{\text{PLCT}}$ is

$$\left\lfloor \bar{M}_{n_p, m_p}^{\text{PLCT}} \right\rfloor_j = \left\lfloor \frac{\bar{M}_{n_p, m_p}^{\text{PLCT}} - d_p(j)}{\mu} \right\rfloor \mu + d_p(j), \quad (14)$$

where $j=0$ or 1 , and $d_p(\cdot)$ is the dither function used as same as that in watermark imbedding process.

Consequently, the extracted watermark bit $\bar{w}_p$ is defined as

$$\bar{w}_p = \operatorname{argmin}_{j \in \{0, 1\}} (|\bar{M}_{n_p, m_p}^{\text{PLCT}}|_j - |\bar{M}_{n_p, m_p}^{\text{PLCT}}|)^2, \quad (15)$$

and then the extracted watermark sequence is $\bar{W} = \{\bar{w}_1, \bar{w}_2, \dots, \bar{w}_{N_{\max}}\}$.

6. Simulation results

In this section, we analyze the capabilities of the PLCT on image representation and image watermarking through simulations. All of the following numerous simulations are executed using double precision floating point values, and all the experiments are conducted on a personal computer having 2.10 GHz Processor with 1.96 GB RAM and Celeron Dual core CPU, and using MATLAB version R2013a under a Windows-XP environment. Meanwhile, we perform all simulations with some normal test images, such as the Elaine image, the Road image, the Boat image, the Man-Woman image and the Lena image, shown in Fig. 1, where Fig. 1(a–d) are gray-scaling images from the well-known benchmark Stirmark, and Fig. 1(f) is a watermark sample

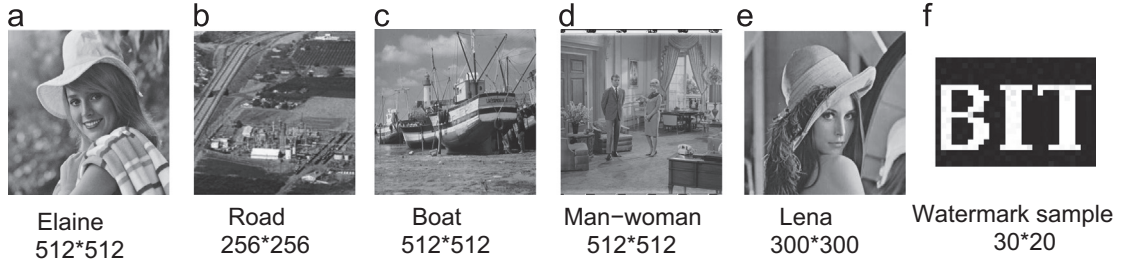


Fig. 1. The test images.

“BIT” viewed as a set W_{BIT} of watermarking bits. It is necessary to exhibit the capabilities of the PLCT compared with some traditional methods. Here, we take the ZMs, the PZMs and the PCET as comparison. The reasons are that (1) the ZMs, the PZMs and the PCET are popular and approved methods applied in the image representation and the image watermarking; and (2) the PCET can be viewed as the special case of the PLCT. In order to examine the capabilities of the PLCT, the followings are organized as five parts: capability on image representation, watermark invisibility, watermark capacity, watermark robustness, and the influence of parameter sensitivity. For a fair comparison, we take the Elaine image as example in the following experiments.

6.1. Capability on image representation

In this subsection, we consider the image representation via the image reconstruction. Fixed $K=50$ and $A = [5.5218 \ 5.5218; 2.5798 \ 2.7609] \in SL(2, \mathbb{R})$, the reconstructed images by four transforms are shown in Fig. 2.

It displays that the ZMs and the PZMs cannot reconstruct the image at the order $K=50$ and the order $K=25$, respectively. While, it is difficult to discriminate between the reconstruction images based on the PCET and the reconstruction images generated by the PLCT with A (short for PLCT_A) through the human naked eyes; hence, the Root-Mean-Square Error (RMSE) is used to measure the difference between an original image and its reconstruction image, defined as

$$\text{RMSE} = \sqrt{\frac{\sum_{x^2+y^2 \leq 1} (f(x,y) - \hat{f}(x,y))^2}{\sum (f(x,y))^2}}. \quad (16)$$

The RMSE values corresponding to Fig. 2 are presented in Table 1 which shows that the PLCT performs slightly better than the PCET on the image representation. It is worth to note that the more the number of transform coefficients are added, the more the reconstructed images are close to the original image.

From the above results, the PLCT has the capability on reconstruction image and less numerical stability issue compared with the ZMs and the PZMs. These results, therefore, motivate us to consider the application of the PLCT in image watermarking.

6.2. Watermark invisibility

To ensure the information safety, the embedded watermark bits should be invisible to human naked eyes; the watermark invisibility is one of the valuable indexes to examine the quality of the transform. And then the peak signal to noise ratio (PSNR) is used as a quantitative index to evaluate the effectiveness of the watermark image f_w on the watermark invisibility, which is defined as

$$\text{PSNR} = 10 \log_{10} \frac{255^2 NM}{\sum_{x=0}^{N-1} \sum_{y=0}^{M-1} (f(x,y) - f_w(x,y))^2}, \quad (17)$$

where f is an image function with size $N \times M$.

Fixed $A_1 = [0.0195 \ 1.9508; -0.5107 \ 0.1951]$, $A_2 = [0.557 \ 5.57; -0.1238 \ 0.557]$, $A_3 = [10.9376 \ 10.9376; 5.3774 \ 5.4688]$, $\mu=0.25$, and the watermark sequence W_{BIT} , Fig. 3 exhibits watermarking images, the corresponding compensation images, and the PSNR values based on the ZMs, the PZMs, the PCET, the PLCT_{A_1} , the PLCT_{A_2} and the PLCT_{A_3} . As observed, the PSNR values determined by the PLCT are bigger than those of the ZMs, the PZMs and the PCET, which implies that the PLCT performs better on the watermark invisibility when the length of the watermark sequence is equal to 600.

Next, it is worth to further consider the PLCT with the order parameters via presenting the relationship between the PSNR values and the N_{max} . In order to analyze it, we fix the quantization step $\mu=0.5$ and some typical parameters shown in Table 2, and then these relationships are exhibited in Table 3 which indicates that the PSNR values are descending along with the increase in N_{max} . It also shows that (1) the factor $A(1,1)/A(1,2)$ affects the performances of the PLCT on the PSNR values; (2) $|A(1,1)/A(1,2)|$ is increasing along with the descending in the PSNR values; and (3) the PLCT with A_3 performs better than the others, where $A_3(1,1)/A_3(1,2)=1$. In a word, the more the embedded watermark bits, the smaller the PSNR values, which leads us to choose the watermarking image satisfying its PSNR value > 40 dB to ensure the watermark invisibility in the rest experiments.

6.3. Watermark capacity

To test the quality of the watermark scheme, the watermark capacity is the other valuable index which is a least upper bound to the watermark length N_{max}

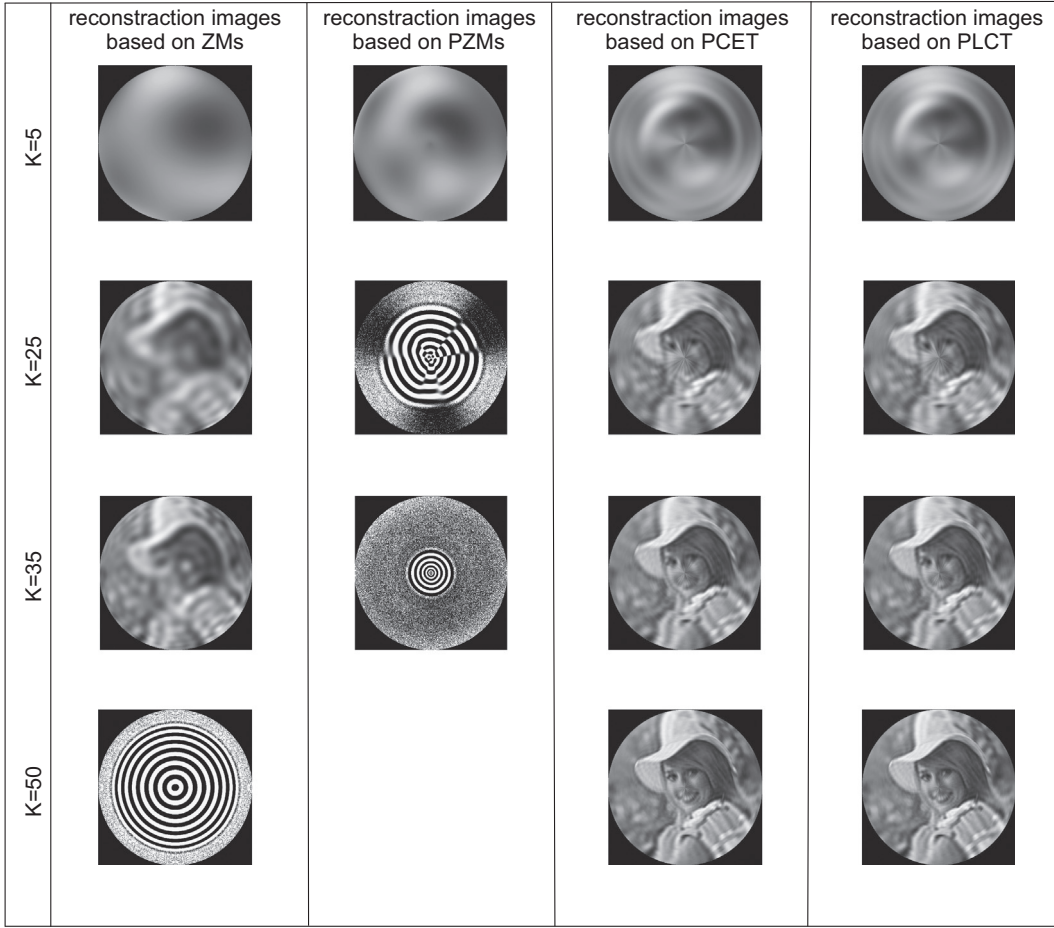


Fig. 2. Reconstruction images based on ZMs, PZMs, PCET and PLCT.

Table 1

Relationship between the RMSE values and the order K .

Methods	K					
	5	15	25	35	45	50
ZMs	0.2469	0.1949	0.1643	0.1365	0.1203	1251.4
PZMs	0.2329	0.1737	653.0	×	×	×
PCET	0.2172	0.1606	0.1265	0.1042	0.0881	0.0830
PLCT _A	0.2172	0.1603	0.1262	0.1040	0.0879	0.0827

discussed in Section 3. The number of the watermarking embedding sets of the ZMs, the PZMs and the PCET is shown as

$$\|S_{ZMs}\| = \begin{cases} \frac{3K^2 + 8K}{16}, & K = 4l, \\ \frac{3K^2 + 10K + 3}{16}, & K = 4l + 1, \\ \frac{3K^2 + 8K + 4}{16}, & K = 4l + 2, \\ \frac{3K^2 + 10K + 7}{16}, & K = 4l + 3, \end{cases}$$

$$\|S_{PZMs}\| = \begin{cases} \frac{3K^2 + 6K}{8}, & K = 4l, \\ \frac{3K^2 + 6K - 1}{8}, & K = 4l + 1, \\ \frac{3K^2 + 6K}{8}, & K = 4l + 2, \\ \frac{3K^2 + 6K + 3}{8}, & K = 4l + 3, \end{cases}$$

and $\|S_{PCET}\| = (3K^2 + 3K)/4$, if $K = 4l(4l + 3)$; otherwise, $\|S_{PCET}\| = (3K^2 + 3K - 2)/4$. It implies that the order K determines the total number of transform coefficients used in image watermarking. Thus, Fig. 4 describes the relationships of the order K and the capacities based on the ZMs, the PZMs, the PCET and the PLCT where the $PLCT_{S^1}$ and the $PLCT_{S^2}$ are determined by Eqs. (9) and (10), respectively. The results indicate that the PLCT with the set S_{PLCT}^1 exhibits better performance on the capacity than that of the others.

6.4. Watermark robustness

The more significant index is the watermark robustness which is used to examine the stabilization of a watermark

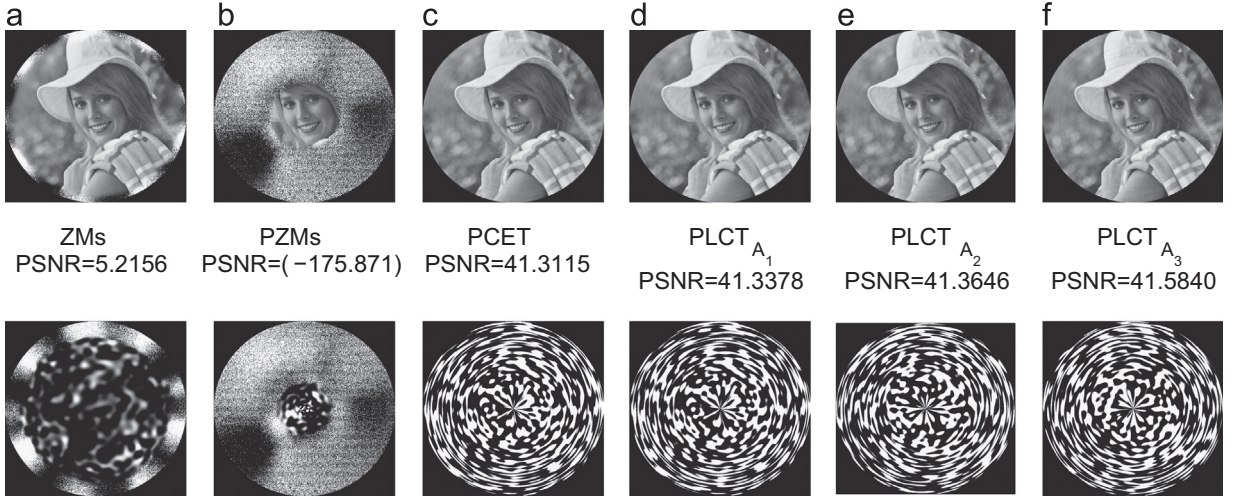


Fig. 3. The first row: the watermarking images and the PSNR values; the second row: the corresponding compensation images.

Table 2

Typical parameters.

i	Parameters of the PLCT	$A_i(1, 1)$ $A_i(1, 2)$
1	$A_1 = [-3.54680.7094; -5.18310.7547]$	-5
2	$A_2 = [-0.27600.2760; -4.30260.6797]$	-1
3	$A_3 = [-0.06550.6551; -1.54280.1626]$	-0.1
4	$A_4 = [0.01190.1190; -8.35370.4984]$	0.1
5	$A_5 = [0.95970.9597; -0.70160.3404]$	1
6	$A_6 = [2.92630.5853; -0.58960.2238]$	5

Table 3

Relationship between PSNR values and the $N_{\max}$.

Methods	$N_{\max}$					
	15	54	99	136	204	256
PLCT _{A₁}	51.153	45.921	43.07	41.659	39.673	38.751
PLCT _{A₂}	49.79	44.532	42.417	41.152	39.673	38.791
PLCT _{A₃}	49.889	44.832	42.836	41.329	39.851	39.07
PCET	50.096	45.499	42.721	41.422	39.912	38.955
PLCT _{A₄}	50.909	45.493	43.208	41.515	39.954	39.185
PLCT _{A₅}	52.443	46.429	43.65	41.86	40.084	39.373
PLCT _{A₆}	50.14	44.502	42.394	41.238	39.835	38.91

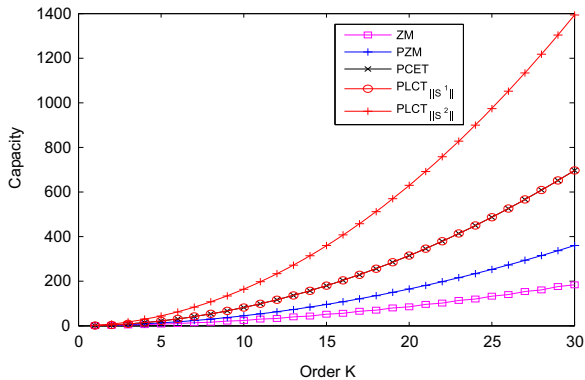


Fig. 4. The relationship between the capacity and the order K .

scheme associated with the transform when the watermark bits are extracted from the watermark image destroyed by various attacks, such as the image rotation, the image scaling, the image compression, and the image noise. To measure the stabilization, the bit error rate (BER) is used as the quantification and defined as

$$BER = \frac{\sum_{i=1}^{N_{\max}} |\bar{w}_i - w_i|}{N_{\max}}, \quad (18)$$

where $\bar{w}_i$ and w_i are the watermark bits belonging to the extracted watermark sequence $\bar{W}_{BIT}$ and in the original watermark sequence W_{BIT} , respectively. The another quantification, the Normalized Cross-Correlation (NC), is

$$NC = \frac{\sum_{n,m} (I_{nm} \bar{I}_{nm})}{\sum_{n,m} (I_{nm})^2}, \quad (19)$$

where I is the watermark sample shown in Fig. 1(f), and $\bar{I}$ is the watermark extraction image determined by $\bar{W}_{BIT}$.

In the following experiments, we take the order K 's of the ZMs, the PZMs, the PCET, and the PLCT as respectively 55, 39, 28, and 28, since the length of the watermark sequence determined by the watermark sample is 600. And the μ is equal to 0.25 for the PSNR values > 40 dB based on the four transforms.

Image rotation robustness: We explore the image rotation robustness when the watermark images are destroyed by rotation attacks. To present the rotation robustness of the PLCT, this part is divided into two categories: First, we consider the relationship between the BER values based on the PLCT and the rotation factors range from 5° to 90° with interval 5° . Fixed $A_1 = [36.2317 \ 18.1158; 3.5680 \ 1.8116]$ and $A_2 = [-11.1399 \ 5.5700; -1.2935 \ 0.5570]$, the results are shown in Fig. 5, where $A_1(1, 1)/A_1(1, 2) = 2$ and $A_2(1, 1)/A_2(1, 2) = -2$. We find that the ZMs and the PZMs have the worse robustness to rotation because of the numerical stability issues, and the PLCT with A_2 performs best among the ZMs, the PZMs, the PCET and the PLCT_{A₁}. To make the results more intuitive, we present the watermark extraction images in Fig. 6. As seen that the PLCT with A_2 performs better than the others under the case of robustness to rotation.

Second, it is necessary to analyze the effect of the order parameters on the BER values. Taken some typical values of $A(1,1)/A(1,2)$ and rotation factors, Table 4 exhibits the relationship between the BER values and the different parameters, where “R” denotes a rotation factor. From Table 4 we have the following conclusions:

- (1) The PLCT with any parameter in $SL(2, \mathbb{R})$ has well performance on the rotation robustness.
- (2) If $A(1,1)/A(1,2) \geq 3$ or $A(1,1)/A(1,2) < -5$, the PLCT with the given parameter performs not better than that of the PCET.

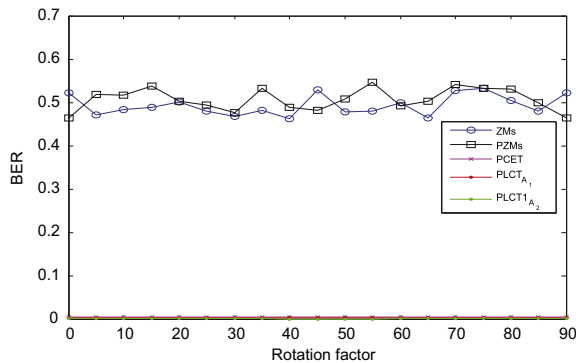


Fig. 5. Performances based on the transforms under watermark robustness to rotation.

- (3) If $|A(1,1)/A(1,2)| < 1$, the performance of the PLCT with the given parameter is similar to that of the PCET.
- (4) If $1 \leq |A(1,1)/A(1,2)| < 3$, the PLCT with the given parameter performs much better.

In short, the PLCT has the well rotation robustness when the watermark bits are extracted from the watermark image destroyed by the rotation attacks.

Image scaling robustness: It is necessary to analyze the image scaling robustness when watermarking images

Table 4

Influence of the PLCT with different parameters on BER values under some rotation attacks.

Methods	$A_i(1,1)/A_i(1,2)$	$R=5^\circ$	$R=15^\circ$	$R=25^\circ$	$R=35^\circ$
PLCT _{A1}	-5	0.0033	0.0033	0.005	0.005
PLCT _{A2}	-3	0.0033	0.0033	0.0033	0.005
PLCT _{A3}	-1	0.0033	0.0033	0.0033	0.0033
PLCT _{A4}	-0.1	0.005	0.005	0.005	0.005
PLCT _{A5}	-0.01	0.005	0.005	0.005	0.005
PCET	0	0.005	0.005	0.005	0.005
PLCT _{A6}	0.01	0.005	0.005	0.005	0.005
PLCT _{A7}	0.1	0.005	0.005	0.005	0.005
PLCT _{A8}	1	0.0033	0.0033	0.0016	0.0016
PLCT _{A9}	2	0.0033	0.0033	0.0033	0.0033
PLCT _{A10}	3	0.01	0.0116	0.0133	0.0116
PLCT _{A11}	5	0.01833	0.0166	0.02	0.0166

	extraction images based on ZMs	extraction images based on PZMs	extraction images based on PCET	extraction images based on PLCT _{A1}	extraction images based on PLCT _{A2}
rot=5	 BER=0.4717 NC=0.3729	 BER=0.5183 NC=0.3242	 BER=0.005 NC=0.9901	 BER=0.0033 NC=0.9934	 BER=0.0017 NC=0.9967
rot=25	 BER=0.48 NC=0.4021	 BER=0.4933 NC=0.3851	 BER=0.005 NC=0.9901	 BER=0.0017 NC=0.9967	 BER=0.0017 NC=0.9967
rot=45	 BER=0.5283 NC=0.3585	 BER=0.4817 NC=0.3585	 BER=0.005 NC=0.9901	 BER=0.0017 NC=0.9967	 BER=0 NC=1

Fig. 6. Watermark extraction images under watermark robustness to rotation.

are destroyed by scaling attacks. To represent the scaling robustness of the PLCT, we first consider the relationship between the BER values and some scaling factors, and then the results are shown in Fig. 7, where $A_1 = [32.5889 \ 16.2945; 3.1975 \ 1.6294]$ and $A_2 = [36.2317 \ 18.1158; 3.5680 \ 1.8116]$. More intuitive results shown in Fig. 8 that

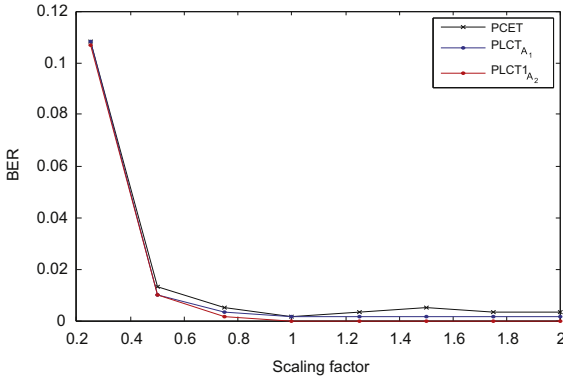


Fig. 7. Performances based on the transforms under watermark robustness to scaling.

the watermark extraction images determined by the PLCT with A_2 are more clear for its smallest NC value compared with the others. Further, we analyze the influences of the PLCT with different parameters on the BER values, and the results are displayed in Table 5. As observed, if

Table 5

Influence of the PLCT with different parameters on the BER values under some scaling attacks.

Methods	$\frac{A_i(1,1)}{A_i(1,2)}$	scal=0.25	scal=0.75	scal=1	scal=1.5
PLCT _{A₁}	−3	0.1183	0.005	0	0.0017
PLCT _{A₂}	−2	0.105	0.005	0.0033	0.0033
PLCT _{A₃}	−1	0.085	0.0067	0.0033	0.005
PLCT _{A₄}	−0.1	0.0983	0.005	0.0033	0.005
PLCT _{A₅}	−0.01	0.1083	0.005	0.0033	0.005
PCET	0	0.1083	0.005	0.0017	0.005
PLCT _{A₆}	0.01	0.1083	0.005	0.0017	0.005
PLCT _{A₇}	0.1	0.115	0.005	0.0017	0.005
PLCT _{A₈}	1	0.1083	0.0033	0.0017	0.0017
PLCT _{A₉}	2	0.1067	0.0017	0	0
PLCT _{A₁₀}	3	0.1033	0.1033	0.0033	0.0033
PLCT _{A₁₁}	5	0.1167	0.01833	0.01667	0.01833

	extracton images based on ZMs	extracton images based on PZMs	extracton images based on PCET	extracton images based on PLCT _{A₁}	extracton images based on PLCT _{A₂}
scal=0.5	 BER=0.5150 NC=0.3289	 BER=0.5167 NC=0.3065	 BER=0.0133 NC=0.9740	 BER=0.01 NC=0.9807	 BER=0.01 NC=0.9807
scal=0.75	 BER=0.525 NC=0.3604	 BER=0.4933 NC=0.3716	 BER=0.005 NC=0.9902	 BER=0.0033 NC=0.9935	 BER=0.0017 NC=0.9967
scal=1	 BER=0.5217 NC=0.3267	 BER=0.4806 NC=0.3655	 BER=0.0017 NC=0.9967	 BER=0.0017 NC=0.9967	 BER=0 NC=1

Fig. 8. Watermark extraction images under watermark robustness to scaling.

$A(1,1)/A(1,2) \geq 3$, the PLCT with the given parameter performs not better than the others; and if $1 \leq A(1,1)/A(1,2) \leq 2$, the BER values based on the PLCT with the given parameter are smaller than those of the others. In a word, the PLCT has the well capability on scaling robustness.

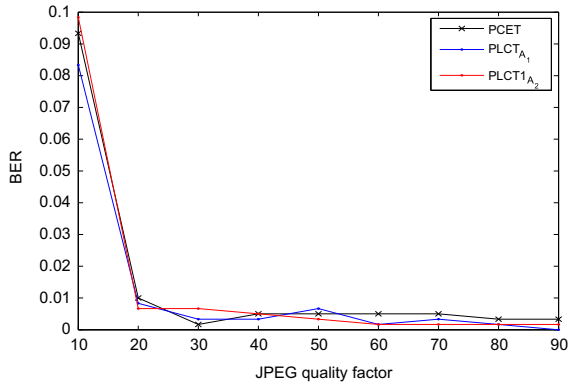


Fig. 9. Performances based on the transforms under watermark robustness to JPEG compression.

Image compression robustness: We analyze the image compression robustness when the watermark bits are extracted from the watermarking images. Firstly, it needs to consider the relationship between the BER values based on the PLCT and the JPEG compression quality factors from 0 to 90 with interval 10, shown in Fig. 9, where $A_1 = [-1.9508 \ 1.9508; -0.7077 \ 0.1951]$ and $A_2 = [36.2317 \ 18.1158; 3.5680 \ 1.8116]$.

The results exhibit that the PLCT with A_2 performs better than the others except for the case of the JPEG quality factor 30. To present more straightforward, we exhibit some watermark extraction images under some JPEG quality factors shown in Fig. 10.

Furthermore, we reveal the effect of the parameter of the PLCT on the BER values. The results are shown in Table 6, where J is the JPEG compression factor for simplicity. It reveals that:

- (1) The PLCT with $A \in SL(2, \mathbb{R})$ has well performance on the JPEG compression attacks.
- (2) If $A(1,1)/A(1,2) \geq 3$ or $A(1,1)/A(1,2) \leq -4$, the PLCT with the given parameter performs not better than that of the PCET.
- (3) If $|A(1,1)/A(1,2)| < 1$, the BER values of the PLCT with the given parameter are similar to that of the PCET.

	extracton images based on ZMs	extracton images based on PZMs	extracton images based on PCET	extracton images based on PLCT $_{A_1}$	extracton images based on PLCT $_{A_2}$
J=20	 BER=0.5050 NC=0.3395	 BER=0.4983 NC=0.3697	 BER=0.01 NC=0.9807	 BER=0.0067 NC=0.9870	 BER=0.0083 NC=0.9841
J=40	 BER=0.5017 NC=0.3226	 BER=0.4783 NC=0.3806	 BER=0.005 NC=0.9902	 BER=0.005 NC=0.9902	 BER=0.0033 NC=0.9935
J=60	 BER=0.5150 NC=0.3466	 BER=0.5083 NC=0.3454	 BER=0.005 NC=0.9902	 BER=0.0017 NC=0.9967	 BER=0.0017 NC=0.9967

Fig. 10. Watermark extraction images under watermark robustness to JPEG compression.

- (4) If $1 \leq |A(1,1)/A(1,2)| \leq 3$, the PLCT with the given parameter performs more better.

Thus, the PLCT performs the stabilization on the JPEG compression robustness when the watermark bits are extracted from the watermark image destroyed by the JPEG compression attacks.

Image noise robustness: At last, we consider the noise robustness of the PLCT when the watermark images are destroyed by the Gaussian noises. We test the capabilities of the PLCT on different levels of noise with standard deviations, such as 5, 9, 13 and 17. The experimental results are shown in Fig. 11 and Table 7.

In addition, for the reported simulation in this paper, we just consider the traditional robustness to rotation, scaling, the JPEG compression, and noise. It is interesting to analyze the robustness based on the PLCT to cropping, nonisotopic scaling, random bending attacks as [8] in future research.

6.5. Influence of parameter sensitivity on the extraction process

In this subsection, we analyze the influence of the PLCT's parameters sensitivity on the BER values and

Table 6
Influence of the PLCT with different parameters on BER values under JPEG compression attacks.

Methods	$A_i(1,1)/A_i(1,2)$	$J=20$	$J=40$	$J=60$	$J=80$
PLCT _{A1}	−3	0.0067	0.0033	0.0017	0
PLCT _{A2}	−2	0.005	0.005	0.0017	0.0033
PLCT _{A3}	−1	0.0067	0.005	0.0017	0.0017
PLCT _{A4}	−0.1	0.0133	0.005	0.0067	0.005
PLCT _{A5}	−0.01	0.0117	0.0067	0.0033	0.005
PCET	0	0.01	0.005	0.005	0.0033
PLCT _{A6}	0.01	0.0117	0.0033	0.0033	0.005
PLCT _{A7}	0.1	0.0083	0.005	0.0067	0.005
PLCT _{A8}	1	0.015	0.0033	0.0033	0.0017
PLCT _{A9}	2	0.015	0.0033	0.0033	0.0017
PLCT _{A10}	3	0.0133	0.0117	0.0067	0.0067
PLCT _{A11}	5	0.03	0.01833	0.0167	0.0167

confirm that the parameter of the PLCT can ensure the safety of the watermark information. In fact, the PLCT with a matrix parameter in $SL(2, \mathbb{R})$ has 3 free parameters compared with the PCET. Thus, it needs to focus on the effect of the PLCT's parameter on the watermark robustness when the watermarking images face some attacks, such as rotation, scaling attacks, and the JPEG compression.

For convenience, we need to define some notations. Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be respectively an embedding process based on the PLCT with parameter A and an extracting process based on the PLCT with parameter B . Then we call $\mathcal{P}_1$ and $\mathcal{P}_2$ consistent (also say $\mathcal{P}_1$ and $\mathcal{P}_2$ have consistency) if A equals to B , otherwise $\mathcal{P}_1$ and $\mathcal{P}_2$ are inconsistent. The $PLCT_{(A,B)}$ is denoted as the PLCT with A in $\mathcal{P}_1$ and B in $\mathcal{P}_2$.

Now, we discuss the effect of parameters of the PLCT on the BER values under the rotation robustness, the scaling robustness and the JPEG compression robustness, which is respectively presented in Figs. 12, 13 and 14, where A, B_1, B_2, B_3, B_4 , are shown in Table 8. We see that performances of the PLCT with B_1, B_2, B_3, B_4 or B_5 in $\mathcal{P}_2$ are worse than that of the PLCT with A as the reason for the inconsistency at $\mathcal{P}_1$ and $\mathcal{P}_2$. While, if ones do not know parameter used in $\mathcal{P}_1$, it is hard to randomly choose parameter as same as A in $\mathcal{P}_2$ because the trouble with parameter is its many varieties in $SL(2, \mathbb{R})$. Therefore, parameters of the PLCT can heavily influence on the BER values when parameters used in $\mathcal{P}_2$ are inconsistency with those in $\mathcal{P}_1$. This infers that the PLCT with parameter has stronger capability to ensure the safety of the watermark information.

Table 7
Average BER for the watermark image under different degrees of noise.

Methods	Attacks					
	1	5	9	13	17	21
ZMs	0.52	0.4967	0.5233	0.5167	0.4817	0.495
PZMs	0.5083	0.475	0.5317	0.5	0.51	0.5383
PCET	0.0026	0.0061	0.0097	0.022	0.0497	0.1228
PLCT	0.0004	0.0057	0.0102	0.020	0.08	0.1

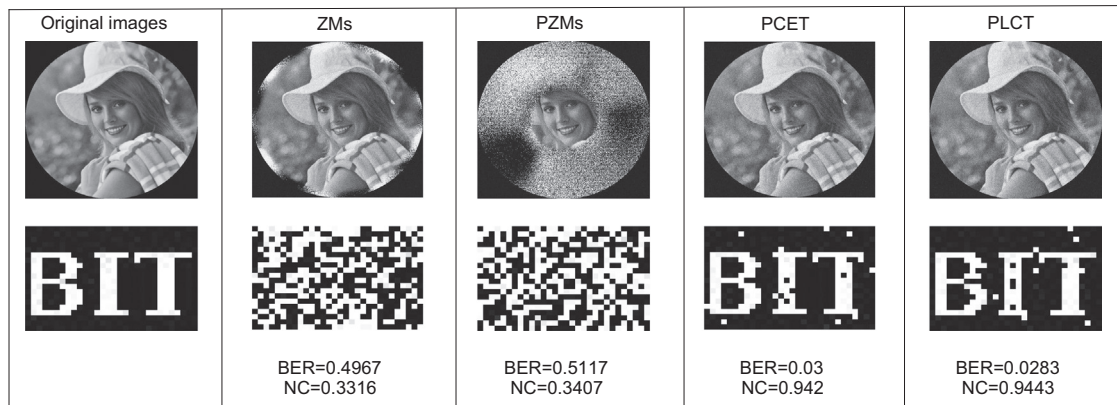


Fig. 11. The first row: the original image and the watermarking images destroyed by noise with standard deviation 13; the second row: the watermark sample and the watermark extraction images.

7. Discussion

Analytically, the computational cost for computing the transform coefficients based on the PLCT for an image of size $N \times N$ is N^2 as same as the special case PCET. While the PLCT has the offset form $\exp\left\{(i/2b)[d(2\pi nb)^2 + ar^4]\right\}$ which affects the radial component and the angle component, thus

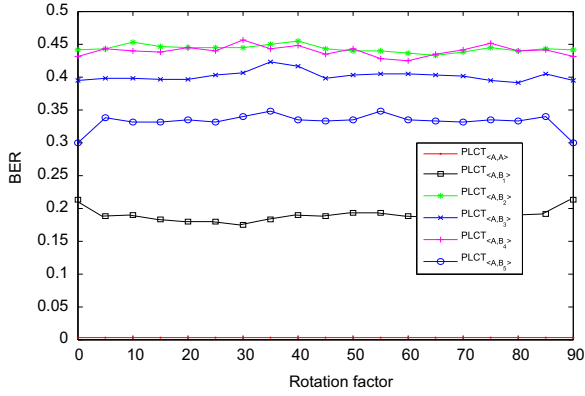


Fig. 12. Effect of parameters of the PLCT1 on the BER values at rotation attacks.

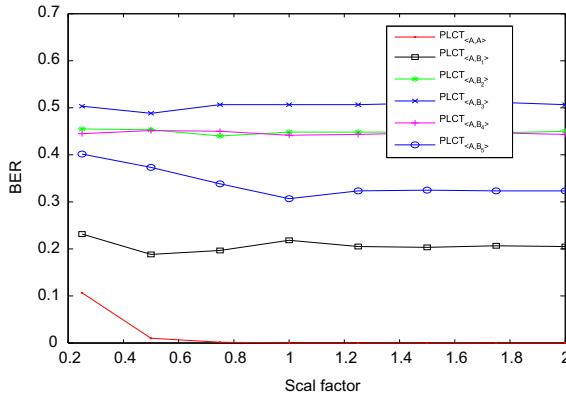


Fig. 13. Effect of parameters of the PLCT1 on the BER values at scaling attacks.

it is crucial to discuss the computational cost of the proposed PLCT compared with the PCET. From the simulations conducted in this paper, all the exponential factors to be used were computed and some useful factors are stored as tables in computer memory in advance for improving the speed. For example, taken the Elaine image with size 512×512 and the watermark sequence with length 128, the times based on the PCET need about 6.5 s and 3.8 s relative to the embedding process and the extraction process, respectively; while the time based on the PLCT is about 6.8 s, and the extraction process requires 3.6 s, because its radial component has more varied form due to the parameter in $SL(2, \mathbb{R})$.

For the above simulations, we have proved that the PLCT with suitable parameters based on the traditional watermarking scheme can perform better than the ZMs, the PZMs and the special case PCET. More important, parameter A is a factor which affects the performance of the PLCT on the image watermarking, meanwhile it can enhance the watermarking information's safety. Up to now, it is not easy to analyze the accurate optimal value of parameter A via the mathematical deduce. While, we have obtained some optimal intervals of parameter A which can make the performance of the PLCT much better on the watermark robustness. In the future researches, we will study the optimization of parameter by the adaptivity watermarking realizations in [12].

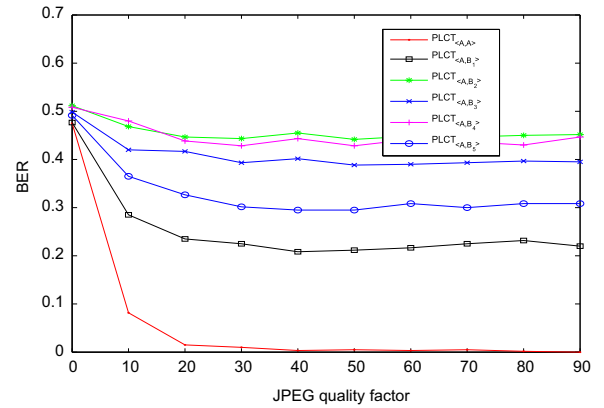


Fig. 14. Effect of parameters of the PLCT1 on the BER values at JPEG compression attacks.

Table 8

The parameters of the PLCT1.

Methods	Parameters used in $\mathcal{P}_1$ and $\mathcal{P}_2$	Features of parameters
$PLCT_{(A,A)}$	$A = \begin{bmatrix} 32.5889 & 16.2945 \\ 3.1975 & 1.6294 \end{bmatrix}, A$	$A \in SL(2, \mathbb{R}), \frac{A(1,1)}{A(1,2)} = 2$
$PLCT_{(A,B_1)}$	$A, B_1 = \begin{bmatrix} 9.7075 & 9.7075 \\ 0.8677 & 0.9708 \end{bmatrix}$	randomly selected $B_1 \in SL(2, \mathbb{R})$
$PLCT_{(A,B_2)}$	$A, B_2 = \begin{bmatrix} 2.5118 & 0.4854 \\ 39.3546 & 8.0028 \end{bmatrix}$	$B_2 \in SL(2, \mathbb{R}), d(A, B_2) = 50.0252$
$PLCT_{(A,B_3)}$	$A, B_3 = \begin{bmatrix} 32.5889 & 8.0028 \\ 5.6529 & 1.4189 \end{bmatrix}$	$B_3 \in SL(2, \mathbb{R}), A(1,1) = B_3(1,1)$
$PLCT_{(A,B_4)}$	$A, B_4 = \begin{bmatrix} 32.5889 & 6.5574 \\ 7.9455 & 1.6294 \end{bmatrix}$	$B_4 \in SL(2, \mathbb{R})$ $A(1,1) = B_4(1,1), A(2,2) = B_4(2,2)$
$PLCT_{(A,B_5)}$	$A, B_5 = \begin{bmatrix} 32.5889 & 9.3399 \\ 29.5208 & 8.4913 \end{bmatrix}$	$B_5 \in SL(2, \mathbb{R}), \frac{B_5(1,1)}{B_5(1,2)} = 2$

8. Conclusion

In this paper, inspired by the methods of traditional moments and the theory of the LCT, we propose the novel polar harmonic transform having parameters in the real special linear group $SL(2, \mathbb{R})$ which is named the polar linear canonical transform (PLCT). It is worthy noting that the approved PCET is the special case of the PLCT when choosing special parameters. By associating the watermark scheme with the PLCT, we evaluate performances of the PLCT on the image reconstruction, the image watermarking, and the influence of parameters sensitivity on the BER values. In a word, experimental results show as follows:

- (1) The PLCT can be applied in the image representation and the image watermarking.
- (2) The PLCT with suitable parameter has the stronger capabilities on the watermark capacity than that of the special case PCET.
- (3) If parameter A satisfies $1 \leq A(1, 1)/A(1, 2) \leq 2$, the PLCT with the given parameter always performs better than the PLCT with the other parameters satisfying the ratio of the $(1, 1)$ -th element to the $(1, 2)$ -th element bigger than 5 or smaller than -5 .
- (4) The effectiveness on the BER can be going bad, when parameters of the PLCT used in the extraction process are inconsistent with those used in the watermark embedding.

In a word, because of the varieties of parameters in $SL(2, \mathbb{R})$, this infers that the watermark security can be improved and guaranteed by the PLCTs.

Acknowledgments

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